

QUEBEC CITY JEAN LESAGE, QC, Canada

WMO: 717080

Lat: 46.8036N

Lon: 71.3817W

Elev: 244

StdP: 14.57

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -14.4	(c) -9.4	(d) -25.0	(e) 1.4	(f) -13.5	(g) -20.1	(h) 1.8	(i) -8.3	(j) 29.8	(k) 19.5	(l) 27.1	(m) 15.2	(n) 9.6	(o) 260	(p) 0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.0	(c) 83.8	(d) 70.0	(e) 81.1	(f) 68.3	(g) 78.4	(h) 66.7	(i) 72.8	(j) 80.3	(k) 70.8	(l) 77.7	(m) 69.0	(n) 75.7	(o) 11.2	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.2	112.3	77.0	68.4	105.1	75.2	66.4	98.1	73.2	36.6	80.7	34.9	77.7	33.3	75.8	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.8	(b) 21.5	(c) 19.1	(e) -21.4	(f) 88.8	(g) 5.1	(h) 2.3	(i) -25.1	(j) 90.5	(k) -28.1	(l) 91.9	(m) -31.0	(n) 93.2	(o) -34.7	(p) 94.9		
(a) 24.8	(b) 21.5	(c) 19.1	(e) -21.9	(f) 76.4	(g) 5.0	(h) 2.7	(i) -25.5	(j) 78.4	(k) -28.4	(l) 79.9	(m) -31.2	(n) 81.5	(o) -34.8	(p) 83.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.2	11.3	14.4	24.8	38.7	53.1	62.4	67.6	65.6	57.6	45.2	32.6	19.4		
	DBStd	21.35	12.60	11.26	10.16	7.57	7.25	6.32	4.92	5.07	6.69	7.03	8.62	10.96		
	HDD50	5036	1199	997	781	346	47	1	0	0	11	182	524	948		
	HDD65	8933	1664	1417	1246	789	375	122	28	55	237	614	973	1413		
	CDD50	1826	0	0	0	6	143	374	545	485	239	33	1	0		
	CDD65	249	0	0	0	0	6	45	108	74	16	0	0	0		
	CDH74	1808	0	0	0	4	108	385	737	460	113	1	0	0		
	CDH80	345	0	0	0	0	20	84	149	75	17	0	0	0		
	WSAvg	8.6	10.1	9.9	9.6	9.3	8.8	7.7	7.2	6.8	7.1	8.4	8.9	9.2		
	PrecAvg	49.8	3.8	3.3	3.6	3.2	4.0	4.5	4.8	4.8	5.0	3.9	4.4	4.6		
	PrecMax	64.9	6.3	7.6	5.9	6.1	7.7	7.2	7.9	8.8	8.6	6.7	7.2	8.7		
(15) Precipitation	PrecMin	38.9	1.7	0.2	0.8	0.9	1.3	1.6	2.2	1.3	1.4	1.5	2.0	1.9		
	PrecStd	5.8	1.4	1.8	1.1	1.1	1.7	1.4	1.5	2.1	1.6	1.3	1.3	1.4		
	DB	40.4	40.6	50.4	68.5	82.8	86.5	87.7	85.5	82.4	70.5	58.7	42.7			
	MCWB	39.1	37.3	41.7	54.2	64.3	71.0	74.1	71.9	70.1	60.9	55.2	41.8			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	36.3	36.6	43.7	60.4	76.8	82.4	83.9	82.0	77.3	64.3	51.5	38.2			
	2% MCWB	34.9	33.9	37.9	48.0	60.6	68.4	70.8	69.3	67.0	57.1	47.5	36.5			
	5% DB	33.1	34.1	40.2	54.6	71.5	78.5	81.0	79.1	72.8	59.6	47.3	35.8			
	5% MCWB	32.0	32.1	35.6	45.1	58.2	65.7	69.0	67.7	64.7	54.4	43.8	34.3			
	10% DB	29.1	31.5	37.4	50.2	66.6	74.8	78.2	76.1	69.2	55.7	43.7	33.6			
	10% MCWB	27.8	29.9	33.7	42.6	55.3	63.9	67.4	66.3	62.5	50.9	41.0	32.2			
	0.4% WB	39.3	37.8	44.2	57.2	68.4	73.8	76.2	74.0	72.4	62.7	55.7	41.7			
	0.4% MCDB	39.7	39.1	47.1	66.4	78.3	82.6	84.7	82.2	80.0	68.1	58.2	42.4			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	35.1	34.3	38.9	49.7	64.1	70.8	73.3	71.6	68.8	58.4	48.5	36.9			
	2% MCDB	36.0	35.9	42.6	57.6	72.5	79.0	80.7	78.2	74.7	63.2	50.3	37.7			
	5% WB	32.1	32.4	36.3	46.4	60.0	68.2	71.1	69.8	66.1	55.1	44.2	34.5			
	5% MCDB	33.3	34.0	39.2	53.2	67.6	75.7	77.9	76.3	71.4	59.1	46.5	35.5			
	10% WB	28.0	29.7	34.2	43.4	56.6	65.7	69.2	68.0	63.4	51.7	41.4	32.5			
	10% MCDB	29.1	31.5	36.9	49.1	65.7	72.4	75.9	74.4	68.2	54.7	43.6	33.5			
	MDBR	15.2	16.4	15.2	15.5	19.0	18.8	18.0	18.1	17.5	14.3	11.8	12.7			
	5% DB	16.0	14.3	16.5	24.7	27.7	24.7	21.9	21.9	21.9	21.1	17.0	11.2			
(36) Mean Daily Temperature Range	MCDBR	16.0	12.9	12.2	15.9	15.6	13.6	11.8	11.8	13.4	14.7	14.1	10.9			
	MCWBR	15.9	13.1	14.4	21.8	23.8	21.0	19.4	18.6	20.0	19.1	15.3	11.2			
	MCWBR	16.0	12.4	11.6	15.3	15.2	13.0	11.6	11.3	13.7	14.5	13.8	11.3			
	MCWBR	16.0	12.4	11.6	15.3	15.2	13.0	11.6	11.3	13.7	14.5	13.8	11.3			
(40) Clear-Sky Solar Irradiance	taub	0.265	0.284	0.325	0.351	0.367	0.403	0.413	0.395	0.370	0.339	0.308	0.292			
	taud	2.311	2.246	2.218	2.346	2.397	2.290	2.282	2.334	2.398	2.479	2.471	2.280			
	Ebn at Noon	264	281	284	287	285	273	269	269	266	259	247	234			
	Edh at Noon	23	31	38	37	36	41	40	37	32	25	21	22			
(44) All-Sky Solar Radiation	RadAvg	445	732	1116	1409	1606	1750	1714	1489	1169	683	395	300			
	RadStd	62	90	91	121	167	111	168	94	99	57	47	30			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+2.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page